

Pablo Scherpereel Arce

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Profile

MSc student at ETH Zurich working on simulation and GNC for multi-spacecraft missions, most recently as sole developer of the end-to-end simulator for a NASA JPL Pre-Phase A study of a formation-flying interferometer at Sun-Earth L2. Now completing a JPL master's thesis on onboard autonomy for the Endurance rover, with a systems engineering background.

Education

ETH Zurich - MSc in Mechanical Engineering
EPFL - BSc in Mechanical Engineering

Sep 2024 - Nov 2026 (expected)
Sep 2021 - Jul 2024

Research Experience

Master Thesis: Endurance Lunar Rover Autonomy Layer, NASA JPL - Pasadena, United States Apr 2026 - Present

Supervisors: Prof. Richard Kornfeld, Prof. Stefan Leutenegger (ETH Zurich) and Dr. Saptarshi Bandyopadhyay (NASA JPL)

- Developing onboard decision-making in Endurance MissionSim, a mission-level Monte Carlo simulation of a 2,600 km lunar south pole traverse, via a receding-horizon scheduler reinterpreting the baseline planner at mission scale.
- Extending the power, data, thermal and fault models and benchmarking Scripted, Reactive and Proactive autonomy in edge-case studies isolating where proactive scheduling pays off, informing JPL's V&V investment in the autonomy stack.
- Lead author of "Understanding the Mission-Level Impact of Onboard Autonomy on the Endurance Lunar Rover," *AIAA SciTech Forum 2027* (extended abstract accepted).

Visiting Student Researcher: Interferometer Mission Study, NASA JPL - Pasadena, United States Oct 2025 - Mar 2026

Supervisors: Rashied Amini and Dr. Saptarshi Bandyopadhyay (NASA JPL)

- Contributed to a Pre-Phase A study of an interferometric constellation for exoplanet imaging and astrometry, two collectors and a beam combiner in formation at Sun-Earth L2, as sole developer of its end-to-end MATLAB simulator.
- Modelled CR3BP halo orbits with lunar perturbation and station-keeping, delay-line control during observations and closed-loop SO(3) attitude tracking under gravity-gradient and SRP torques with thruster allocation.
- Built field-of-regard and Δv -optimized schedulers and traded delay-line tolerance and baseline against Δv , deriving preliminary L1/L2 requirements on relative navigation accuracy, control authority and metrology precision.

Engineering Experience

Project Manager & Chief Engineer: "Hermes" Ethanol/LOx Rocket, ARIS - Zurich, Switzerland Aug 2024 - Oct 2025

- Led the 50+ member team from MDR to FRR with ECSS-inspired requirements, interface and configuration control.
- Served as Assembly Lead for propulsion, propellant supply and avionics during launch-site operations.

Recovery Team Lead & Structural Engineer: "Nordend" & "Firehorn", ERT - Lausanne, Switzerland Sep 2022 - Aug 2024

- Led a 6-person team developing the Firehorn dual-event recovery subsystem with reefing-line pyrotechnic cutters.
- Designed the Nordend payload structure and Firehorn engine bay using launch-load FEA and topology optimization.

Projects

Semester Project: Cold-Gas Reaction Control System, ETHZ - Zurich, Switzerland

- Designed a 3-axis cold-gas RCS to CDR state of maturity for a 100-km suborbital rocket, sizing the nozzles and nitrogen supply system and implementing a bang-bang PD controller under a phased state machine with fault-triggered abort.
- Verified it against a full requirements matrix with a verification method per requirement, through 6-DOF Simulink runs with disturbance, actuator delay and sensor noise injected, and structural FEA on ascent and chute-deployment loads.

Bachelor Project: Supersonic Wind Tunnel, EPFL - Lausanne, Switzerland

- Designed the test section, pressure vessel and diffuser of an indraft Mach 1.5-3 wind tunnel, taken through CDR.

Skills

GNC & Dynamics: orbital mechanics (CR3BP), formation keeping, attitude control, disturbance modelling, 6-DOF simulation

Simulation & Analysis: MATLAB/Simulink, Python scientific stack, C, Monte Carlo methods, FEA and CFD (Abaqus, ANSYS)

Systems Engineering: ECSS, requirements & V&V, verification matrices, trade studies, Pre-Phase A studies, MDR to FRR

Languages: French (Native), Spanish (Native), English (C2), German (A1)